SOFTWARE TEST PLAN

CampusConnect: University Club, Event & Activity Management Platform

Attribute	Specification Detail
Document Identifier	CC-STP-2026-V1.0 (ISO/IEC/IEEE 29119-3:2013 Standard)
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Reviewing Body	Faculty Academic Review Board | Division of Computing and Systems
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1. Introduction and Test Objectives

1.1 Purpose

This Software Test Plan (STP) defines the verification and validation framework for CampusConnect, a centralized university extracurricular and club management platform. This document operationalizes the acceptance criteria defined in the Software Requirements Specification (CC-SRS-2026-V2.0) and the technical architecture specified in the Software Design Document (CC-SDD-2026-V1.0).

1.2 Test Objectives

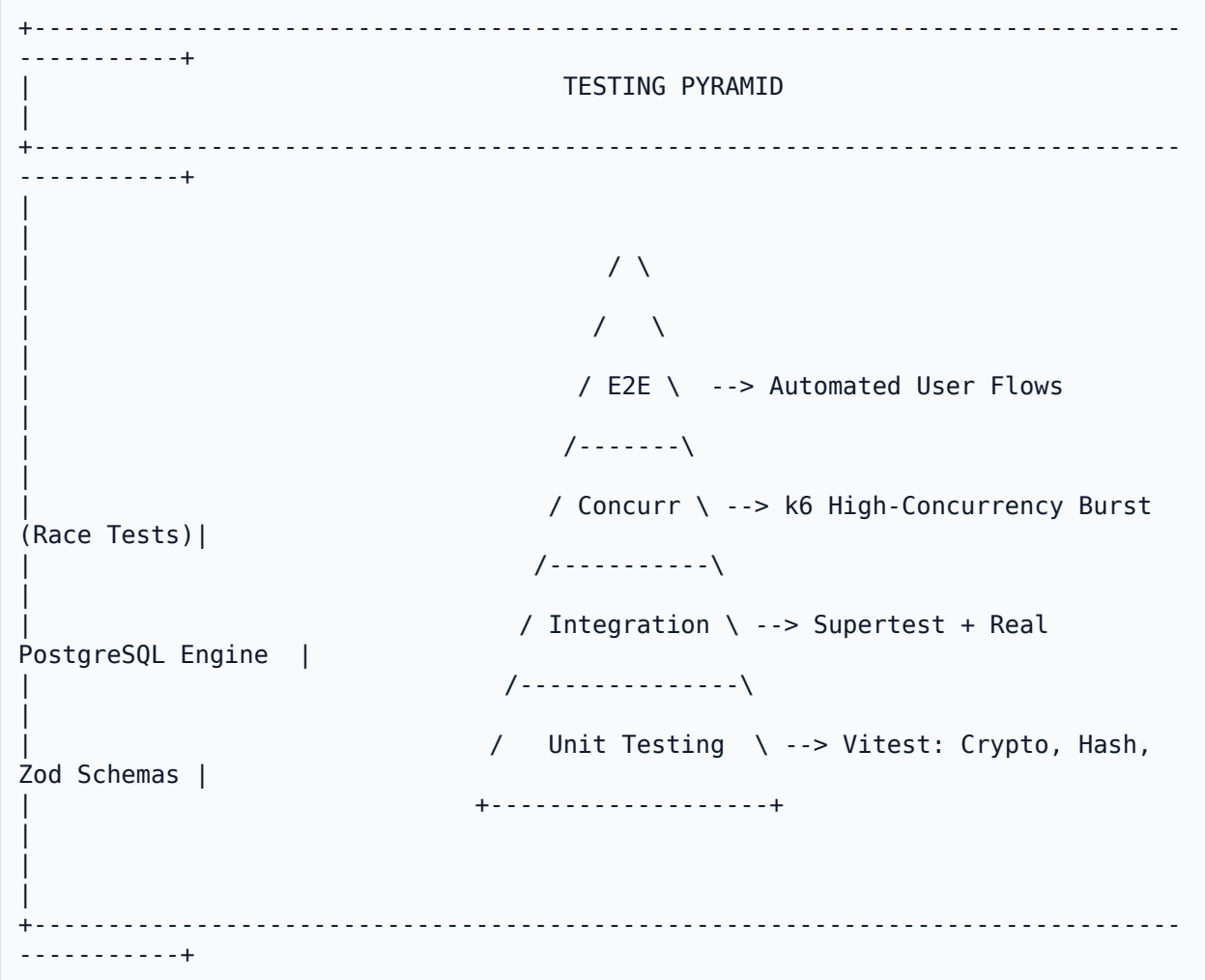
The testing program must verify system correctness, security posture, and resilience across the following core dimensions:

- **1. ACID Concurrency:** Validate that parallel registration bursts hitting capacity limits never produce over-booking, negative slot counts, or unhandled database deadlocks.
- **2. Anti-Proxy Attendance:** Verify that rotating HMAC-SHA256 tokens strictly expire within their designated time window (20 seconds) and resist screenshot-sharing replay attacks.
- **3. Digital Certificate Integrity:** Confirm that generated PDF certificates accurately reflect event participation records and that any payload tampering invalidates the verification hash.
- **4. Access Control Enforcement:** Validate that horizontal and vertical privilege escalation attempts between Student, Club Executive, Faculty Advisor, and System Administrator roles are blocked with HTTP 403 Forbidden.

- **5. Coverage and Performance:** Enforce a minimum of 85% statement coverage across business logic modules and confirm p95 API response times under 250ms for read operations and under 400ms for write operations.

2. Test Strategy and Testing Pyramid

CampusConnect adopts a multi-tier test pyramid balancing fast unit tests, deterministic integration tests, high-concurrency race condition simulations, and security penetration suites:



1. Level 1 - Unit Testing (Vitest): Enforces >= 85% statement coverage on pure business logic, input validation schemas, token rotating math, and cryptographic hash verification.

2. Level 2 - API Integration Testing (Supertest + PostgreSQL): Exercises full HTTP lifecycle across authentication, role-based authorization, and transactional database rollbacks.

3. Level 3 - Concurrency Stress Testing (k6): Validates ACID row-level locking by dispatching 100+ virtual users simultaneously against an event with 5 remaining slots.

4. Level 4 - Security & Penetration Testing: Audits OWASP Top 10 defenses, expired token rejection, certificate forgery detection, and rate limiting barriers.

3. Test Environment and Tooling Configuration

Testing Tier	Primary Tooling	Configuration & Scope
Unit Test Runner	Vitest 3.0+	Fast in-memory TypeScript runner with V8 coverage.
HTTP Integration	Supertest 7.0+	Direct route invocations against Express application.
Test Relational DB	PostgreSQL 16 (Docker)	Isolated campusconnect_test database on port 5433.
Cache & Rate Limiting	Sliding Window / Redis 7	In-memory sliding window map with Redis 7 container configuration.
Load & Stress Engine	k6	Virtual user scripting for race-condition validation.
Coverage Enforcer	Vitest Coverage	Enforces 85% statement and 80% branch thresholds.

4. Comprehensive Test Case Specifications

4.1 Authentication and Access Control (TC-AUTH)

Test ID	Test Case Title	Precondition	Input / Trigger	Expected Result
TC-AUTH-01	Student Registration	Valid email format	POST /api/v1/auth/register	HTTP 201 Created; bcrypt hash persisted
TC-AUTH-02	Duplicate Email Block	Email already exists	POST /api/v1/auth/register	HTTP 409 Conflict; account collision blocked
TC-AUTH-03	Valid Credential Login	Account ACTIVE	POST /api/v1/auth/login	HTTP 200 OK; JWT returned, cookie set
TC-AUTH-04	Invalid Password Block	Registered user	POST /api/v1/auth/login (bad pass)	HTTP 401 Unauthorized; enumeration blocked
TC-AUTH-05	Rate Limit Enforcement	Zero attempts prior	6 failed logins in 60s	HTTP 429 Too Many Requests on 6th call
TC-AUTH-06	Token Refresh Rotation	Valid refresh cookie	POST /api/v1/auth/refresh	HTTP 200 OK; new JWT and rotated refresh token

4.2 Event Governance and Concurrency (TC-EVENT)

Test ID	Test Case Title	Precondition	Input / Trigger	Expected Result
TC-EVENT-01	Draft Event Creation	Club Exec authenticated	POST /api/v1/events (valid body)	HTTP 201 Created; initial status DRAFT
TC-EVENT-02	Advisor Review Approval	Event in SUBMITTED state	POST /api/v1/events/:id/review (APPROVE)	HTTP 200 OK; status transitions to PUBLISHED
TC-EVENT-03	Advisor Rejection Notes	Event in SUBMITTED state	POST /api/v1/events/:id/review (REJECT)	HTTP 200 OK; rejection_reason logged
TC-EVENT-04	Capacity Race Condition	Capacity=5, registered=4	10 concurrent registrations	Exactly 1 CONFIRMED; 9 WAITLISTED
TC-EVENT-05	Waitlist FIFO Promotion	Capacity=50, 3 on waitlist	POST /api/v1/registrations/:id/cancel	HTTP 200 OK; first waitlisted promoted

4.3 Dynamic Attendance and Anti-Proxy Testing (TC-ATTEND)

Test ID	Test Case Title	Precondition	Input / Trigger	Expected Result
TC-ATTEND-01	Live Rotating QR Mint	Club Exec for event	GET /api/v1/events/:id/attendance/live-qr	HTTP 200 OK; HMAC token with 20s TTL
TC-ATTEND-02	Valid Check-in Scan	Confirmed participant	POST /api/v1/events/:id/attendance/scan	HTTP 200 OK; check-in timestamp saved
TC-ATTEND-03	Expired Token Block	Token generated 60s ago	POST /api/v1/events/:id/attendance/scan	HTTP 400 Bad Request; expired token rejected
TC-ATTEND-04	Duplicate Scan Block	Already checked in	POST /api/v1/events/:id/attendance/scan	HTTP 409 Conflict; duplicate check-in denied
TC-ATTEND-05	Unregistered Scan Block	No registration record	POST /api/v1/events/:id/attendance/scan	HTTP 403 Forbidden; registration required

4.4 Digital Certificates and Verification (TC-CERT)

Test ID	Test Case Title	Precondition	Input / Trigger	Expected Result
TC-CERT-01	Eligible Certificate Mint	Attendance verified	GET /api/v1/certificates/me	HTTP 200 OK; returns signed serial

Test ID	Test Case Title	Precondition	Input / Trigger	Expected Result
TC-CERT-02	Ineligible Certificate Block	No attendance record	GET /api/v1/certificates/me	HTTP 403 Forbidden; attendance mandatory
TC-CERT-03	Public Credential Verification	Valid serial in DB	GET /api/v1/certificates/verify/:serialNumber	HTTP 200 OK; returns valid details (no login)
TC-CERT-04	Tampered Serial Block	Unknown serial string	GET /api/v1/certificates/verify/:serialNumber	HTTP 404 Not Found; error envelope

5. Concurrency Stress Specification (k6 Script)

The concurrency test suite simulates sudden registration rushes using k6. 100 virtual users attempt simultaneous registration against an event with 10 open slots:

```
import http from 'k6/http';
import { check } from 'k6';

export const options = {
  vus: 100,
  iterations: 100,
  thresholds: {
    http_req_duration: ['p(95)<400'],
    http_req_failed: ['rate<0.01'],
  },
};

export default function () {
  const res =
http.post('http://localhost:5000/api/v1/events/EVENT_ID/register');
  check(res, {
    'Valid Response': (r) => r.status === 201 || r.status === 200,
    'No Server Failures': (r) => r.status !== 500,
  });
}
```

6. Pass / Fail Criteria and Quality Gates

1. Unit Test Pass Rate: 100% of unit tests in Vitest must pass without skips.

2. Code Coverage: Global statement coverage >= 85%, branch coverage >= 80%.

3. Integration Testing: 100% of Supertest route tests against PostgreSQL must pass.

4. Concurrency Assertion: Zero over-allocation and zero deadlocks during k6 load tests.

5. Security Audits: Zero High or Critical security vulnerabilities in dependency tree.